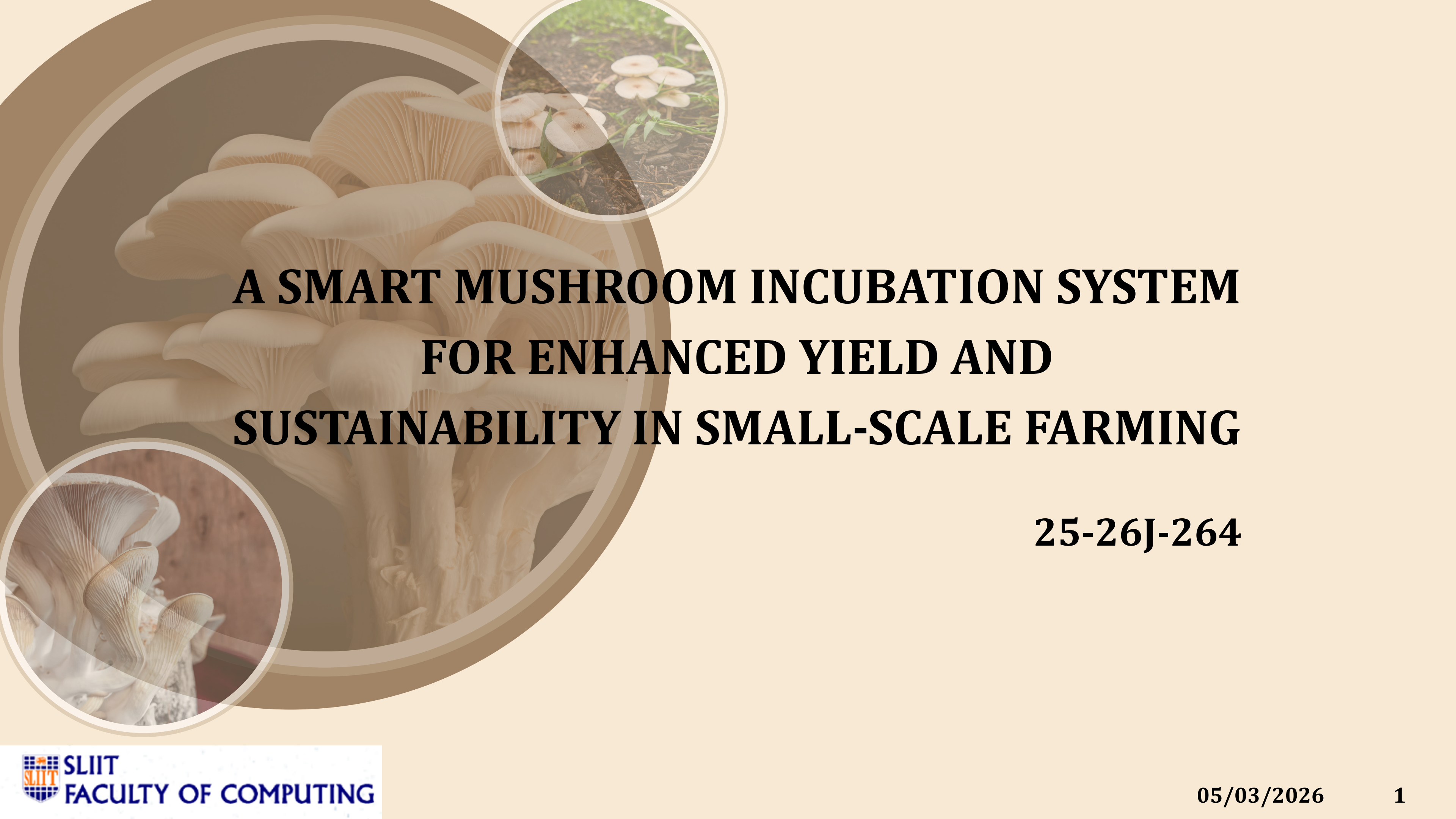




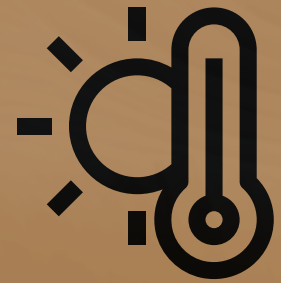
# **A SMART MUSHROOM INCUBATION SYSTEM FOR ENHANCED YIELD AND SUSTAINABILITY IN SMALL-SCALE FARMING**

**25-26J-264**





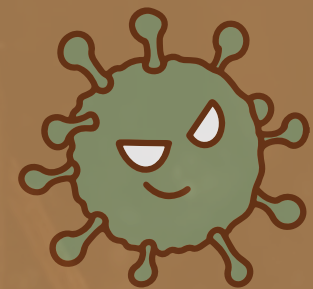
# Problem and motivation



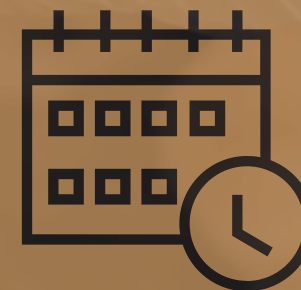
Manual environment  
monitoring



Late pest detection



Contamination risks



No fruiting  
prediction

# Our Solution

AI + IoT system for smart mushroom monitoring

- ➔ Real-time environment monitoring
- ➔ AI-based detection
- ➔ Predictive farming insights
- ➔ Mobile dashboard alerts



# Key features

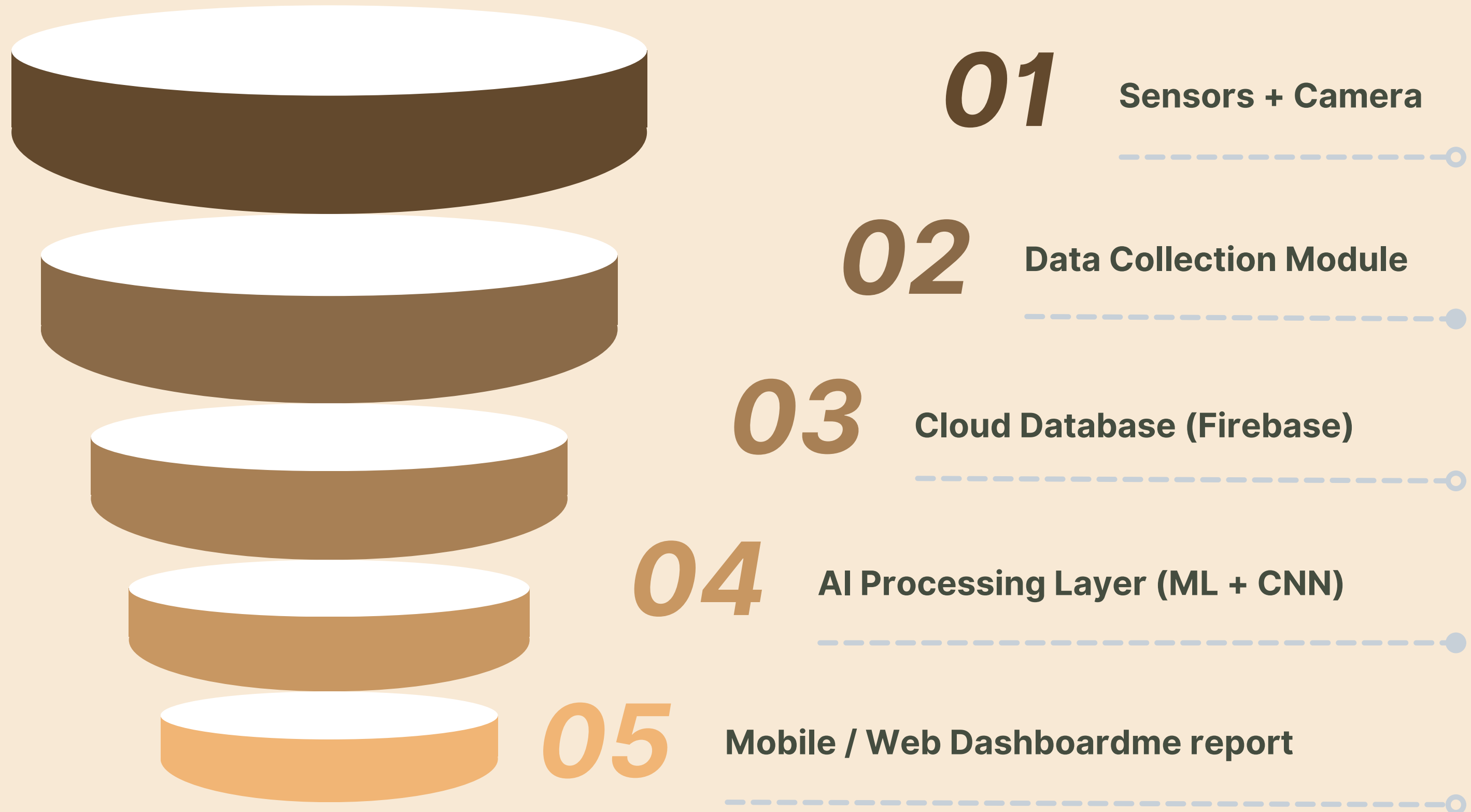
## Functional features

- FR1 – Environmental monitoring
- FR2 – Pest & contamination detection
- FR3 – Fruiting time prediction

## System quality

- Performance
- Usability
- Reliability
- Security

# System Architecture





| Progress at PP1 - 50%                                                                                                                                                                                                           | Progress at PP2 - 90%                                                                                                                                                                                                                                                       | In progress - 10%                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Literature review</li> <li>• System architecture design</li> <li>• Initial sensor data collection</li> <li>• Image dataset preparation</li> <li>• Basic UI dashboard design</li> </ul> | <ul style="list-style-type: none"> <li>• IoT sensor integration</li> <li>• CNN models for pest &amp; contamination detection</li> <li>• ML model for fruiting prediction</li> <li>• Mobile &amp; web dashboard development</li> <li>• Firebase cloud integration</li> </ul> | <ul style="list-style-type: none"> <li>• Model Optimization and Performance Improvement</li> <li>• Implementation of Real-Time Alert System</li> <li>• Pilot Testing with Farmers</li> <li>• Final System Validation</li> <li>• Finalizing the Incubation Prototype</li> </ul> |

# Farmer Perspective & Commercial Value



No need for manual  
rack checking

Early alerts prevent  
contamination

I can focus on other  
daily work

Sinhala mobile app  
is easy to use



# Cost–Benefit Analysis



## SYSTEM INVESTMENT

*Robot Arm  
Mechanical + Rail & Sensors  
Installation*



## LABOR COST

*Daily worker wage – LKR 1,800  
Monthly labor cost – LKR 54,000*

**System Investment  $\approx$  LKR 260,000**

## SAVINGS

*Monthly savings – LKR 54,000  
Annual savings – LKR 648,000*



## PAYBACK PERIOD

*~5 Months*



# Demonstration time





**IT22063182 | M L Vidanaarachchi**

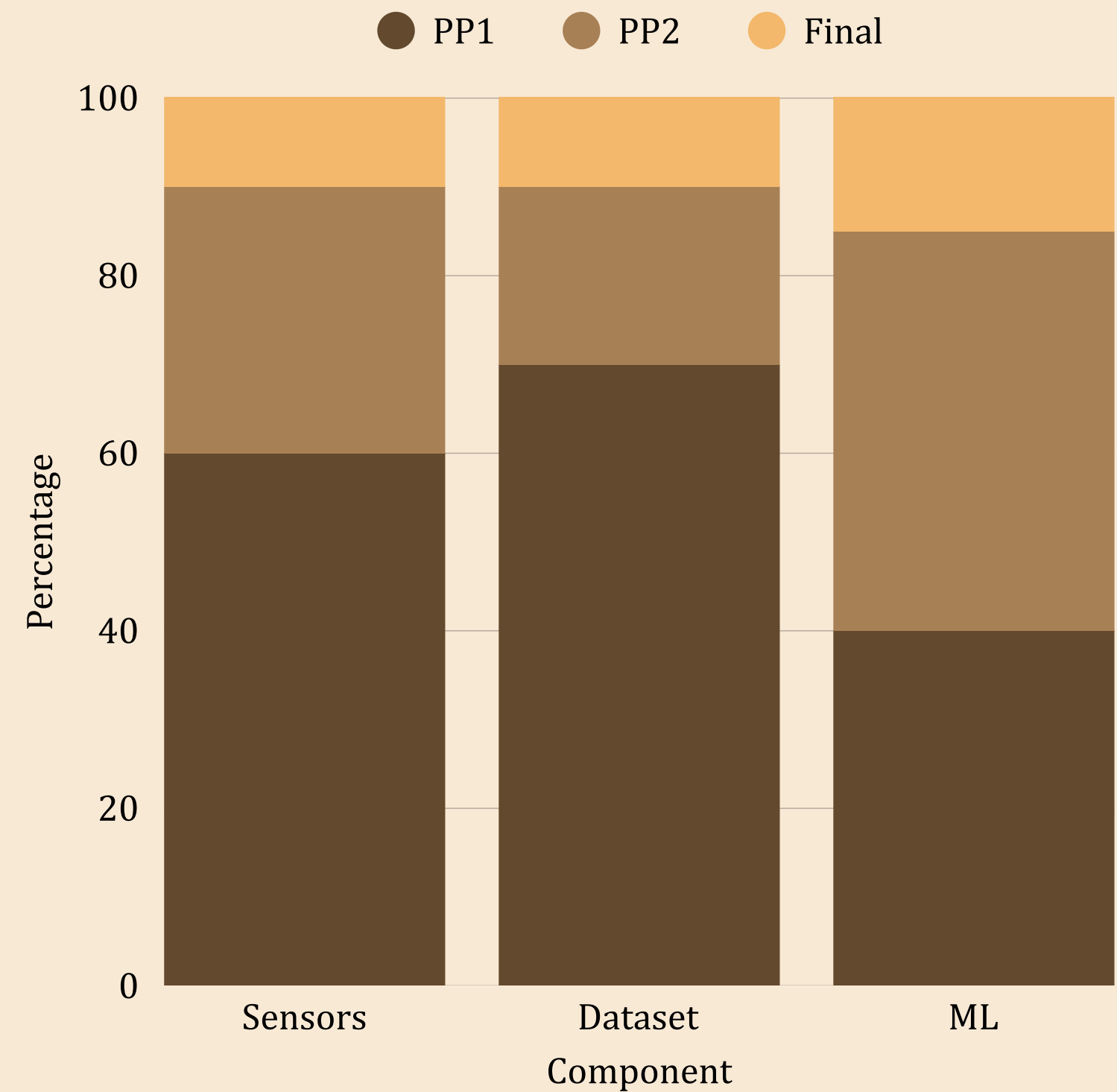


## **Predictive Fruiting Module using Environmental Sensing and Machine Learning**

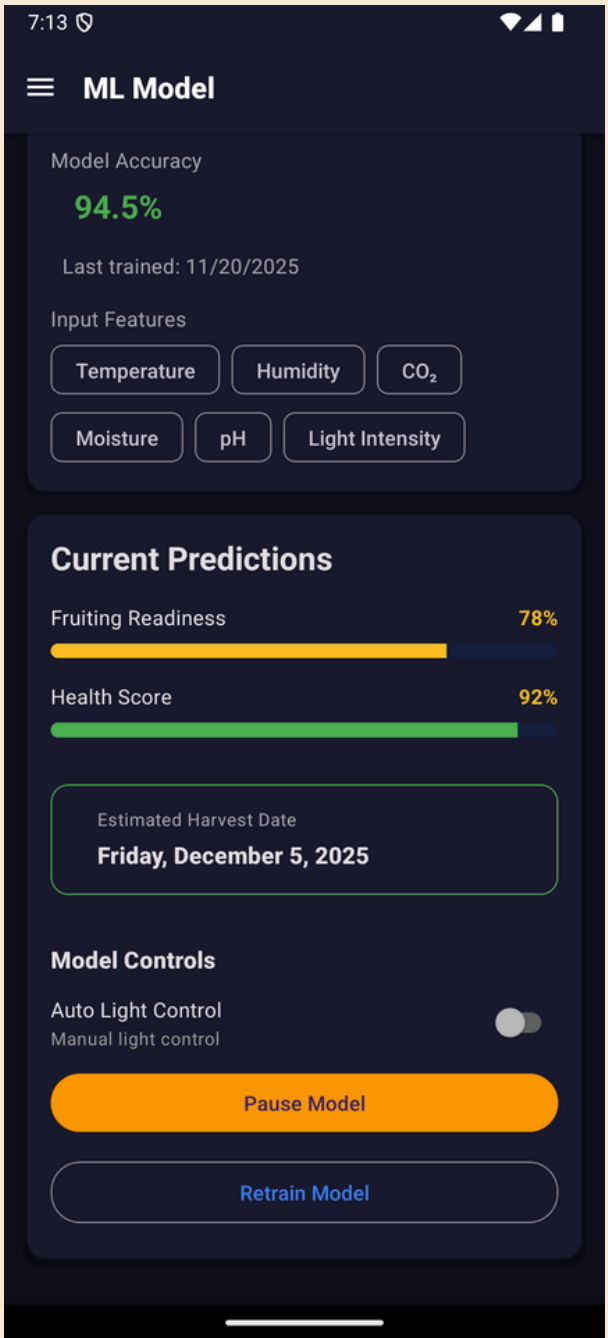
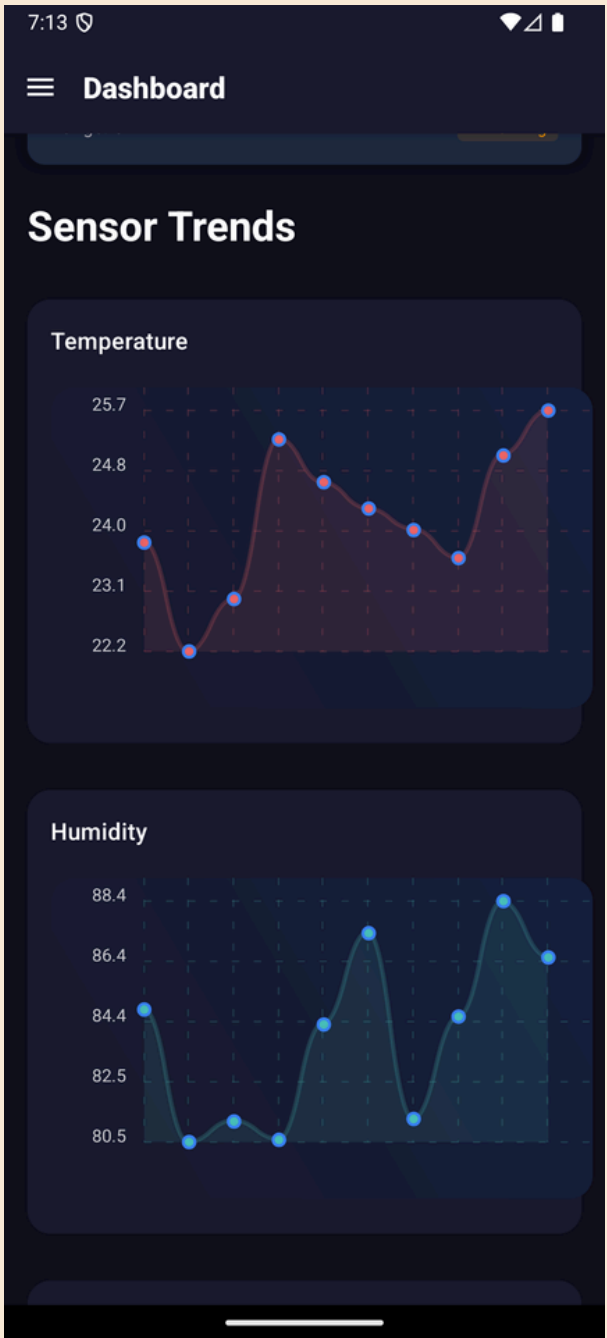
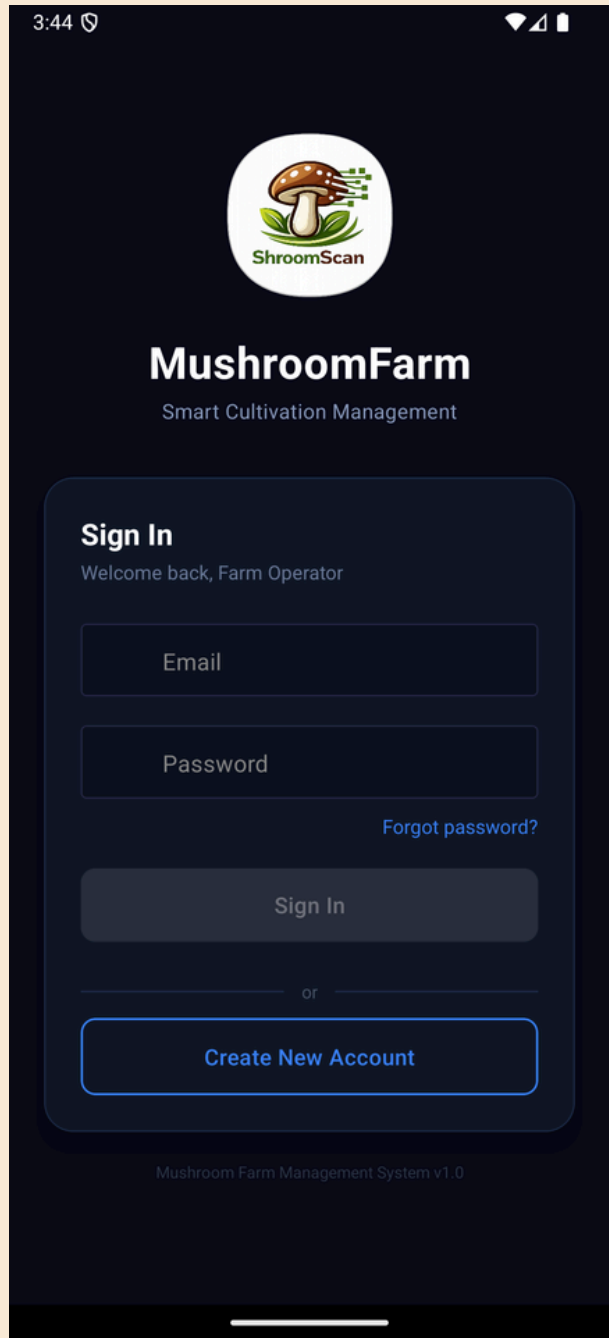
Specialization: Information Technology



# Predictive Fruiting Module using Environmental Sensing and Machine Learning



# Predictive Fruiting Module using Environmental Sensing and Machine Learning





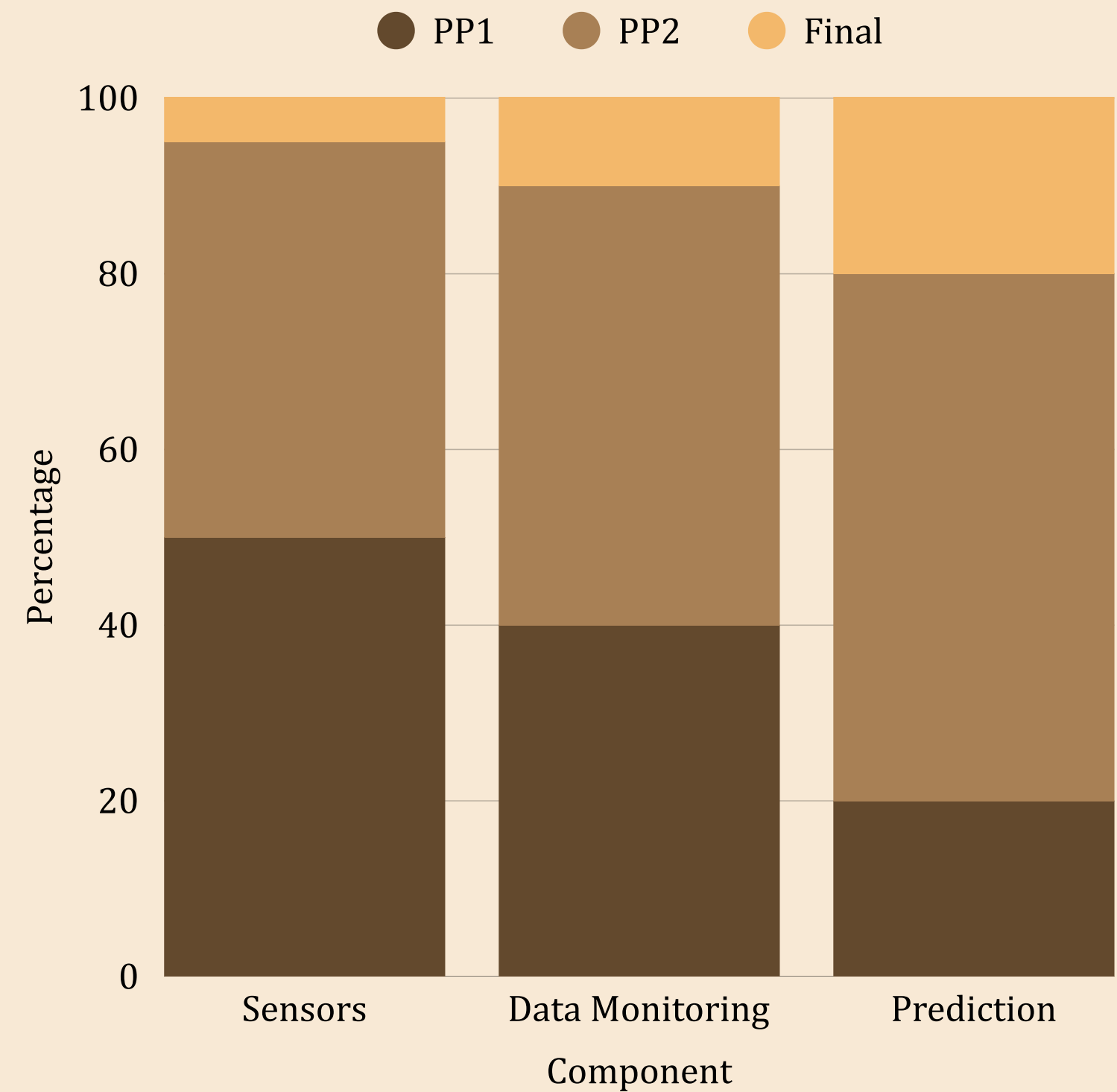
**IT22586384 | S W O P K Waduge**



# **Predictive Advisory System for Moisture and pH Management in Mushroom Cultivation**

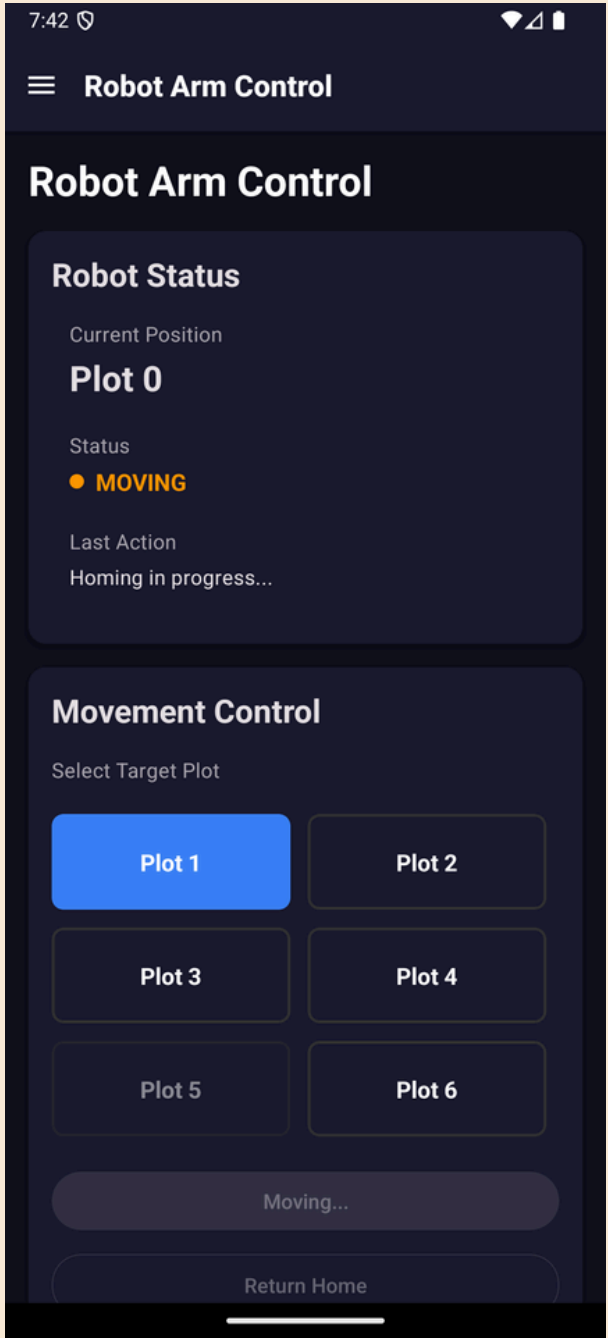
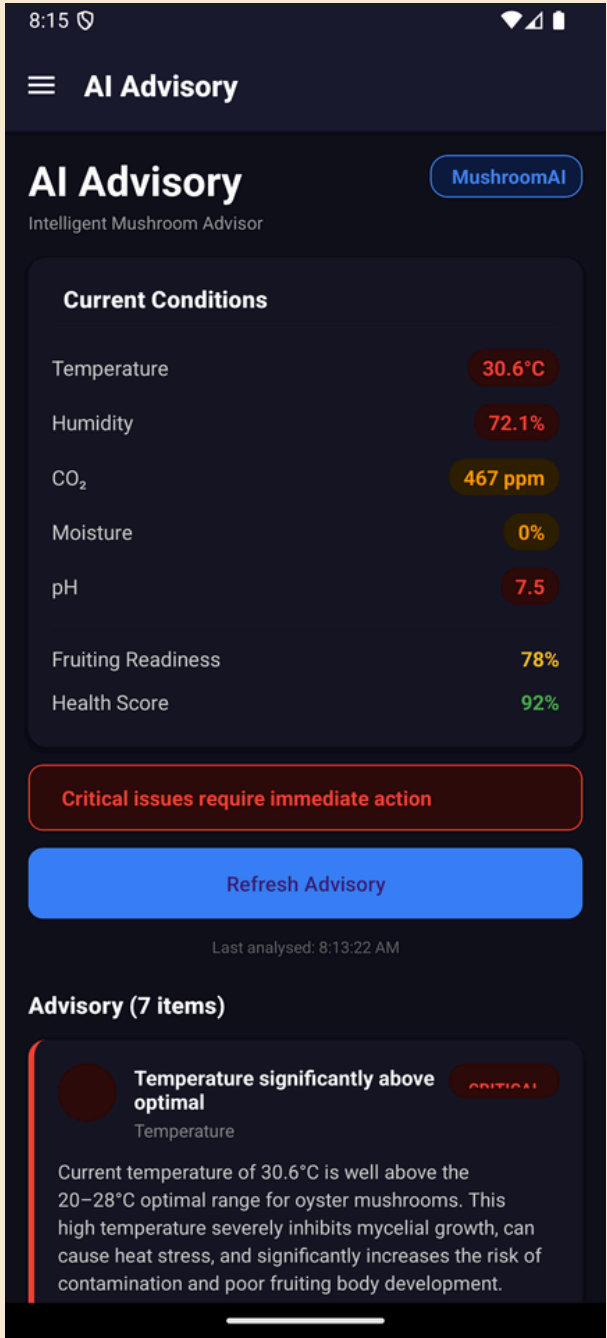
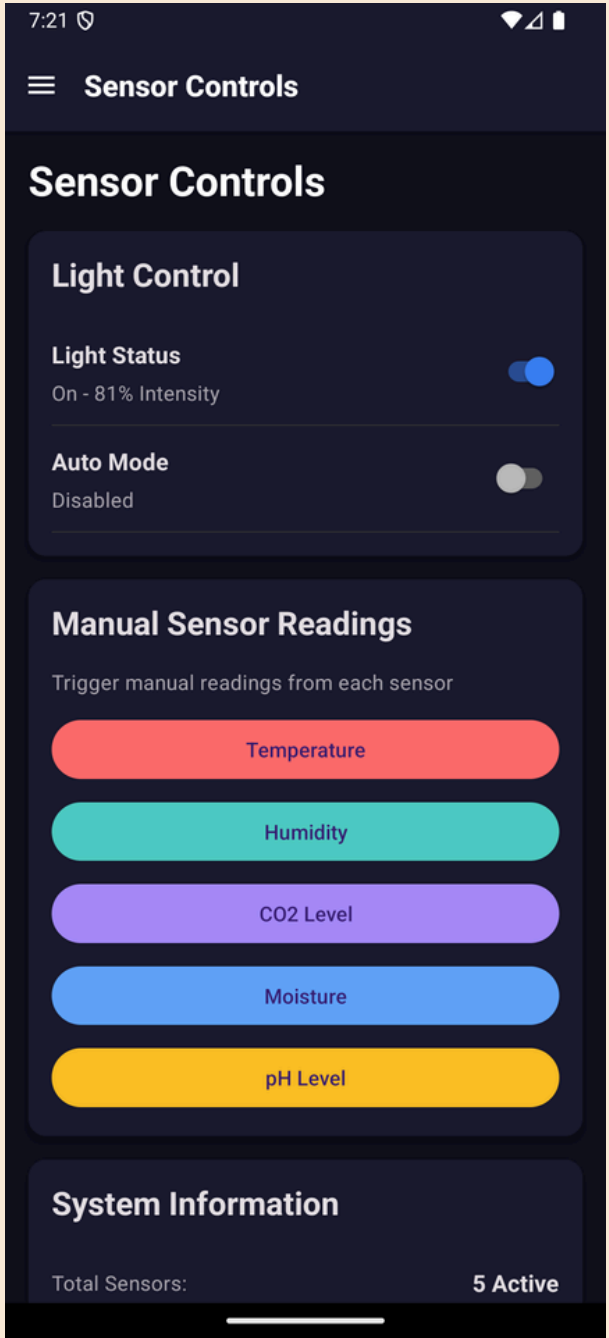
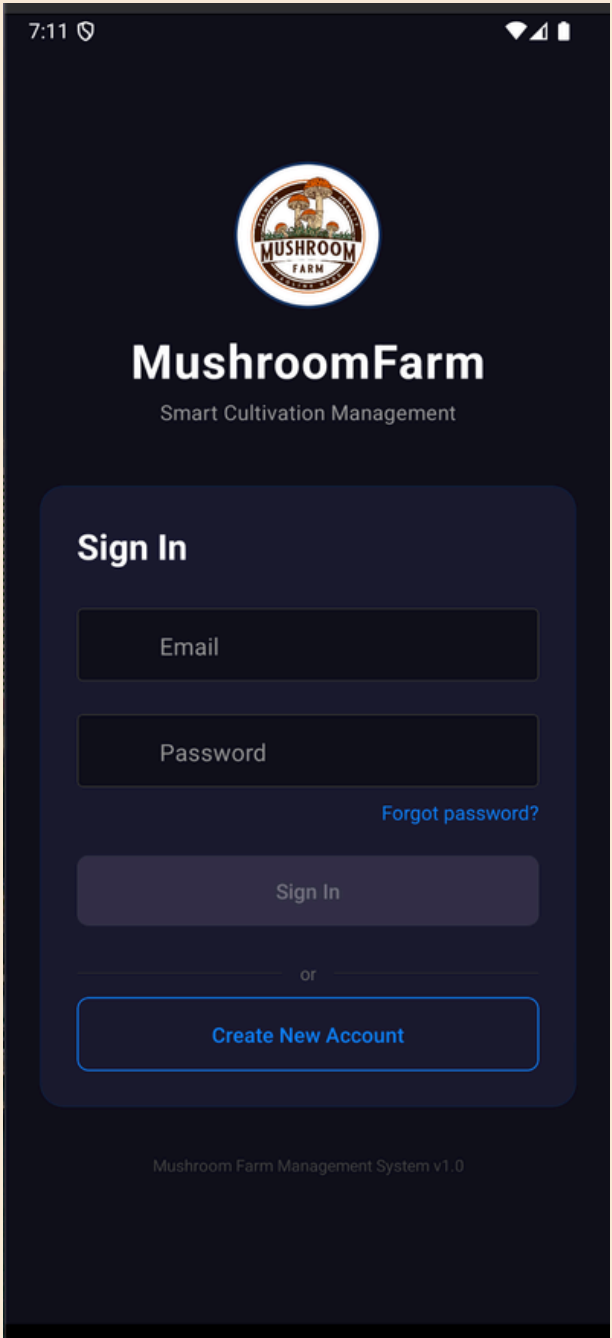
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# Predictive Advisory System for Moisture and pH Management in Mushroom Cultivation





# Predictive Advisory System for Moisture and pH Management in Mushroom Cultivation



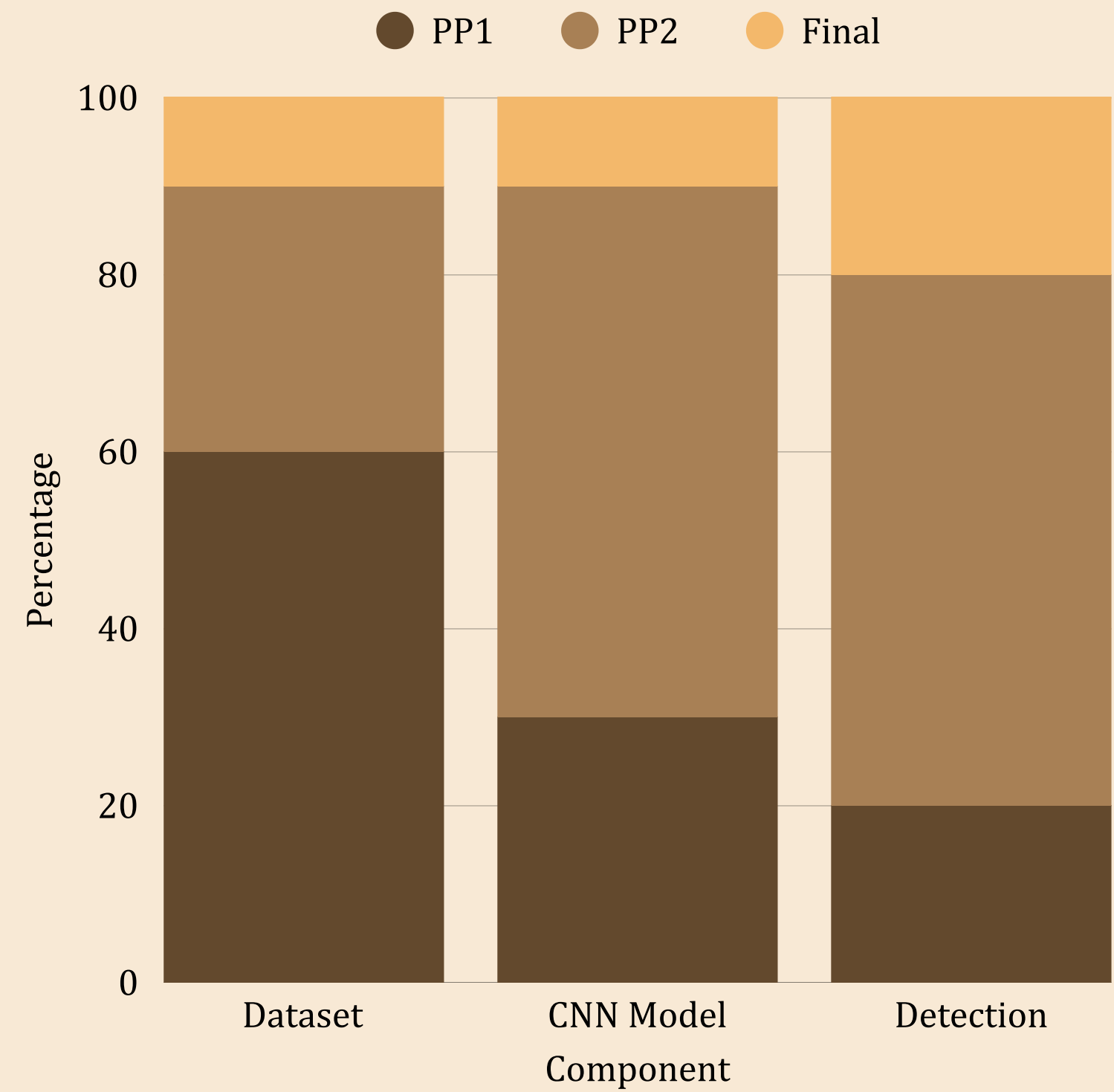
**IT22290960 | B H A S Peiris**



# **Fungal Contamination Detection Module using Image Processing and Deep Learning**

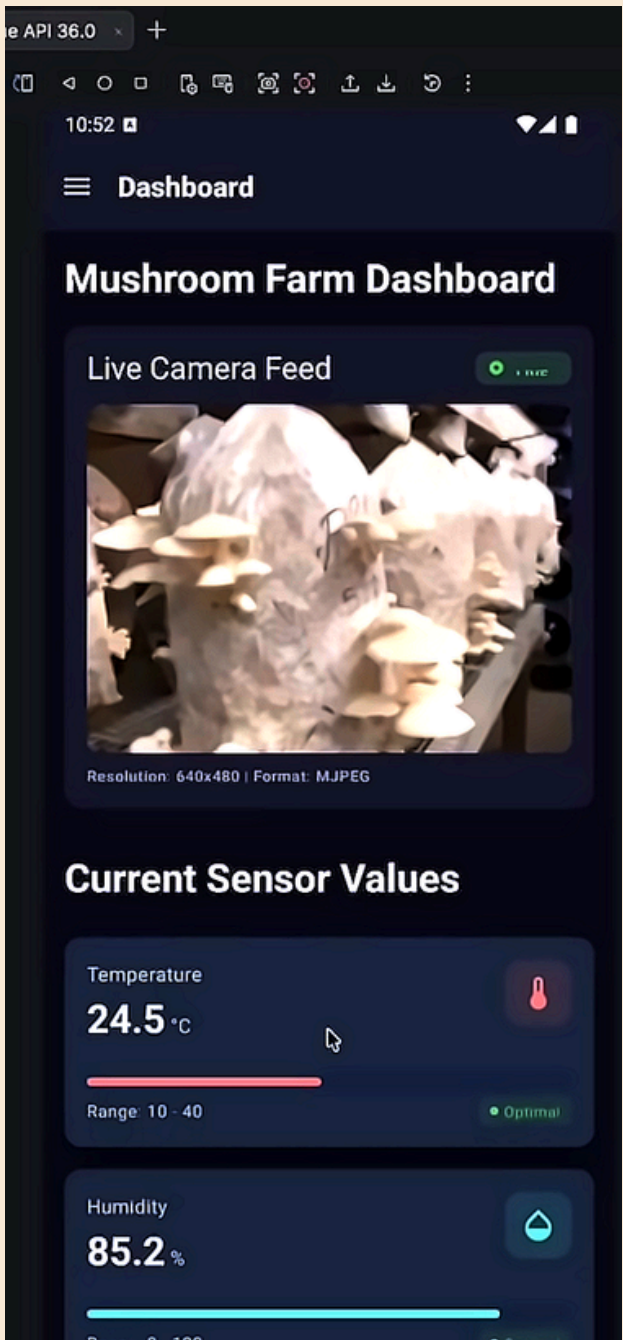
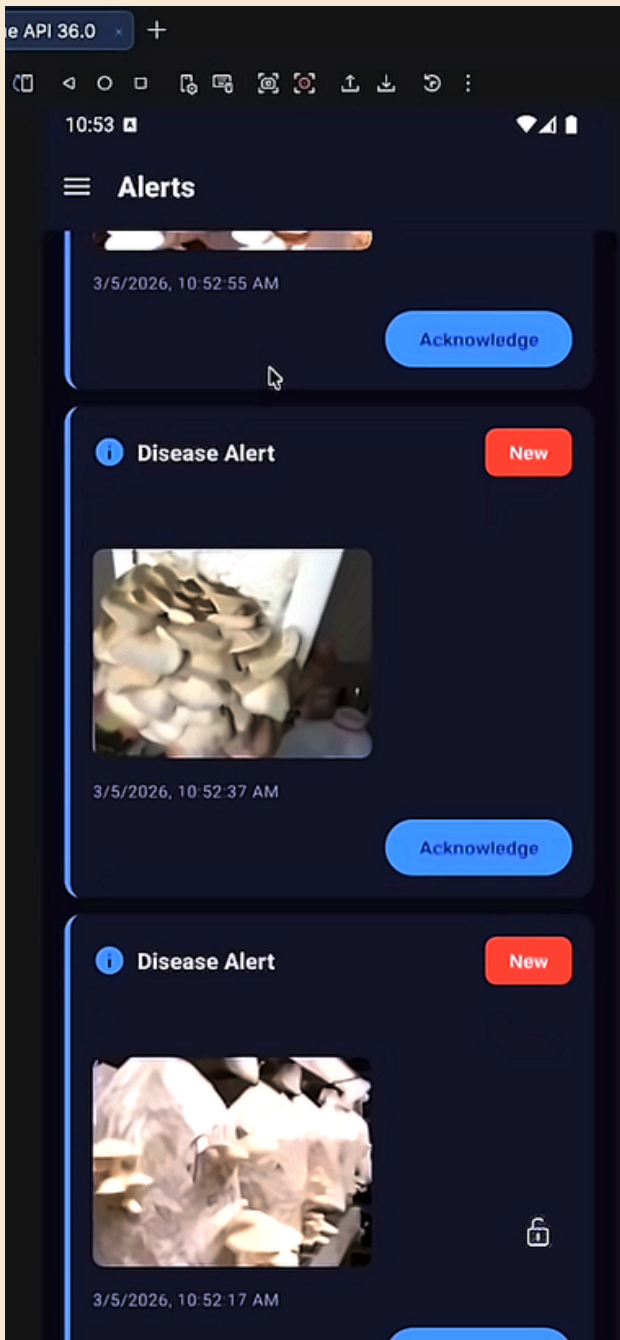
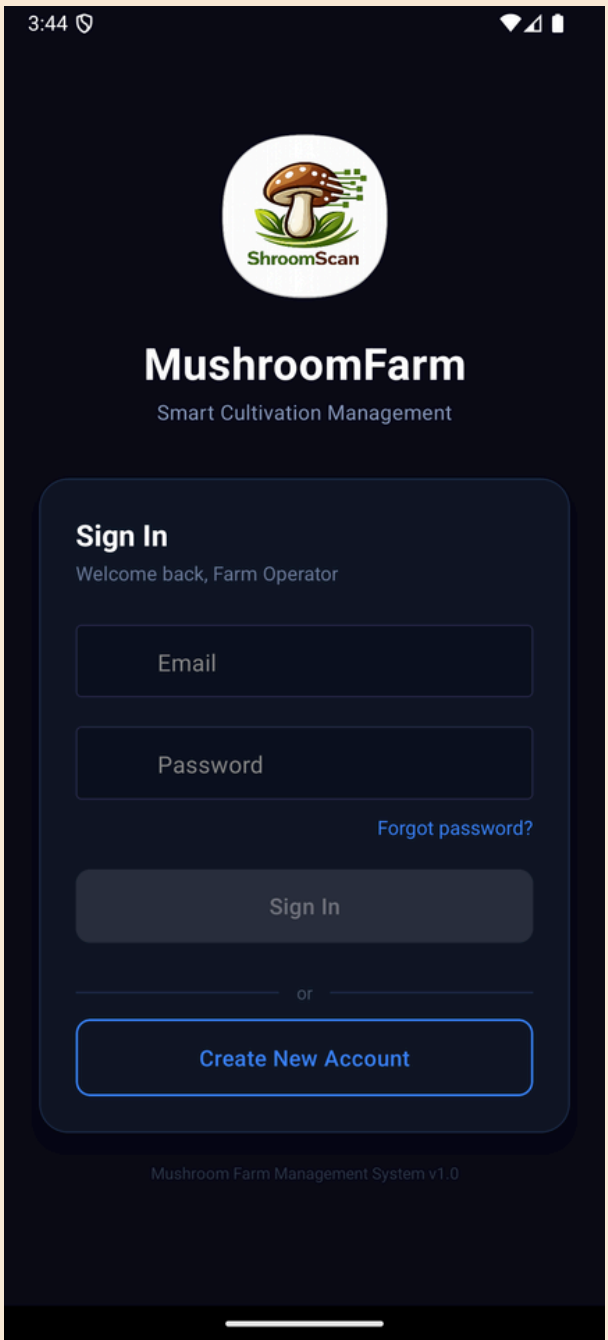
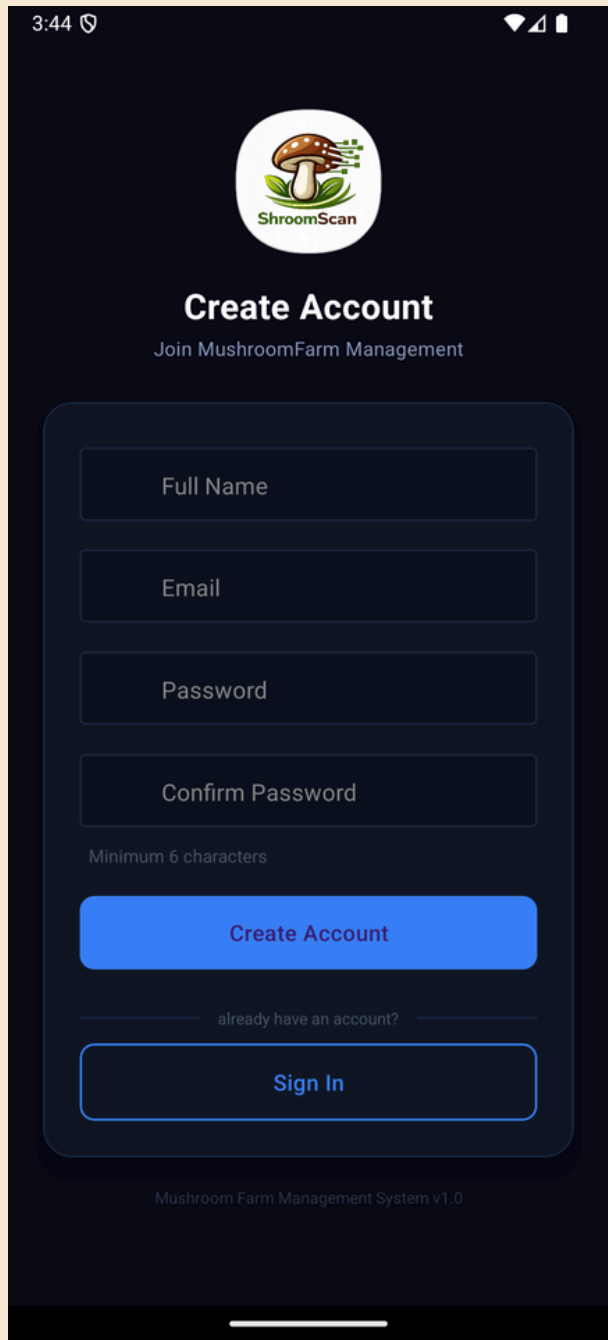
Specialization: Information Technology

# Fungal Contamination Detection Module using Image Processing and Deep Learning





# Fungal Contamination Detection Module using Image Processing and Deep Learning



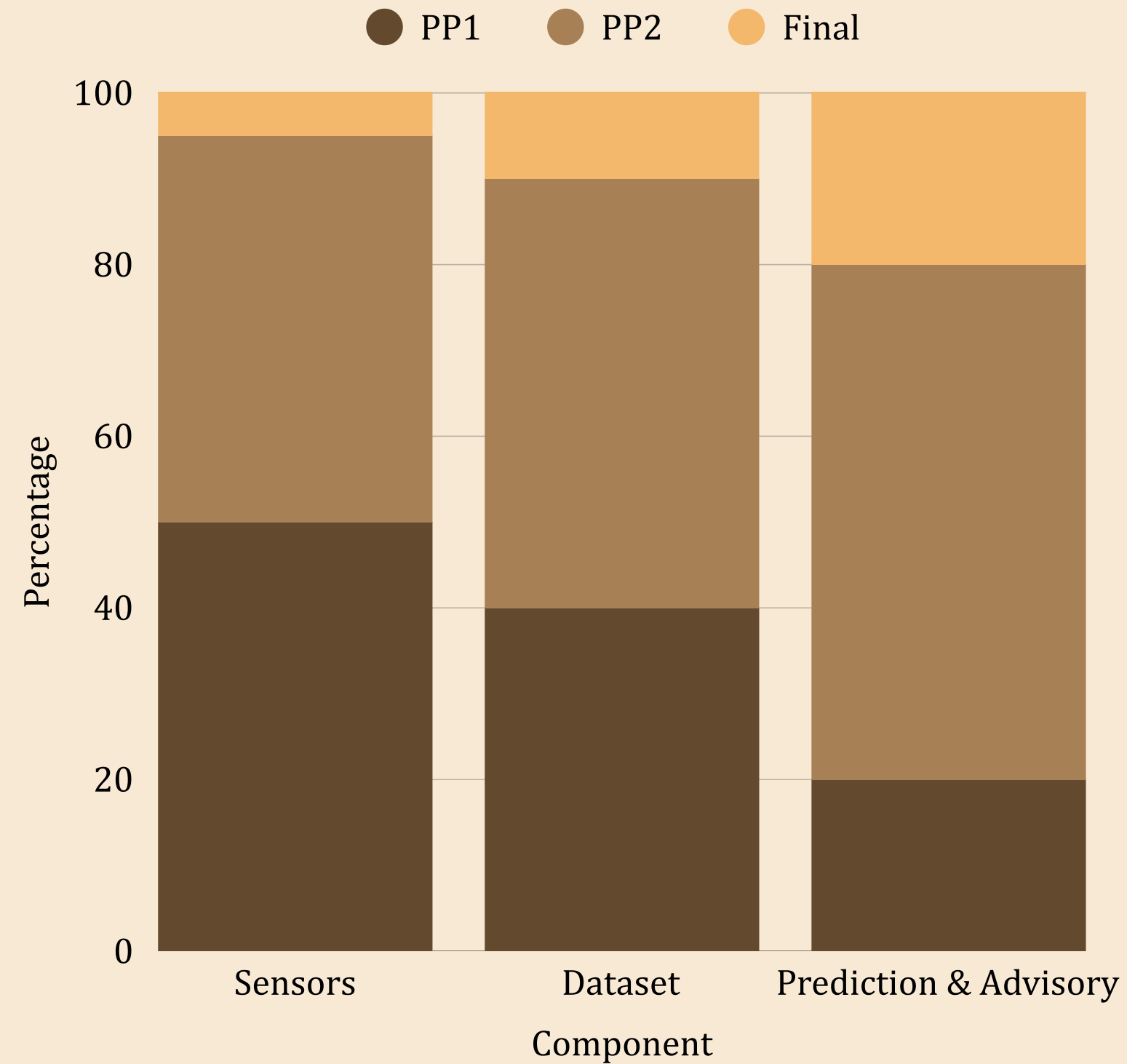
**IT22062888 | G A I W Wijesinghe**



## **Pest Management Module using Real-Time Detection and Automated Response**

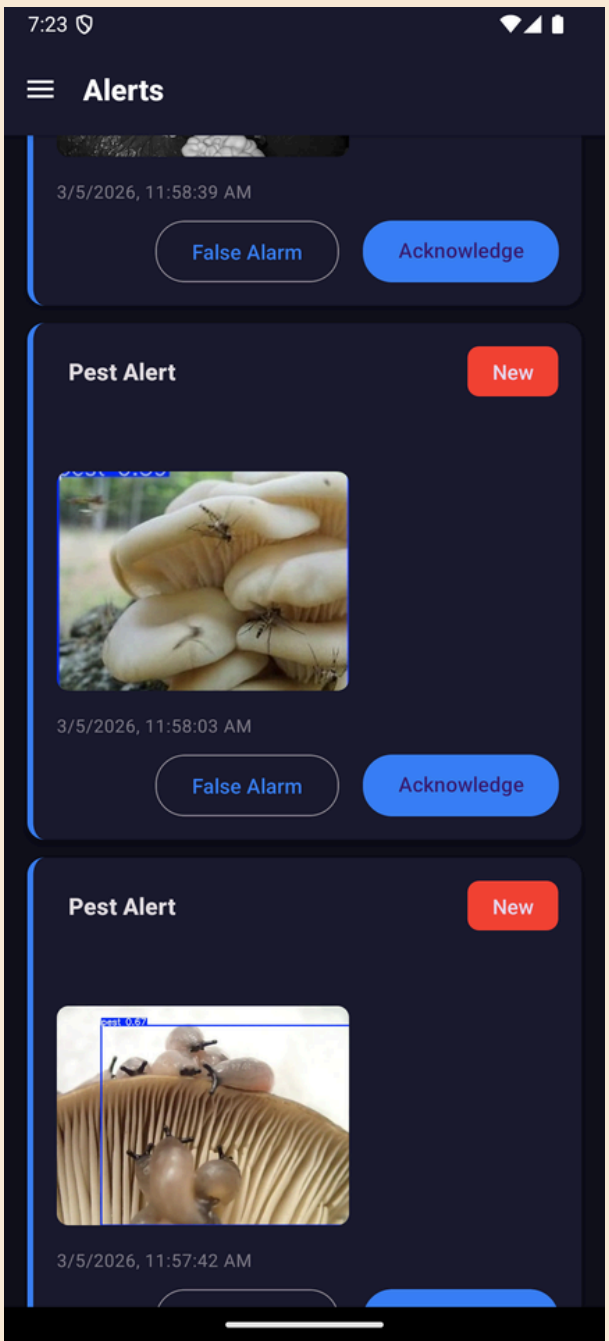
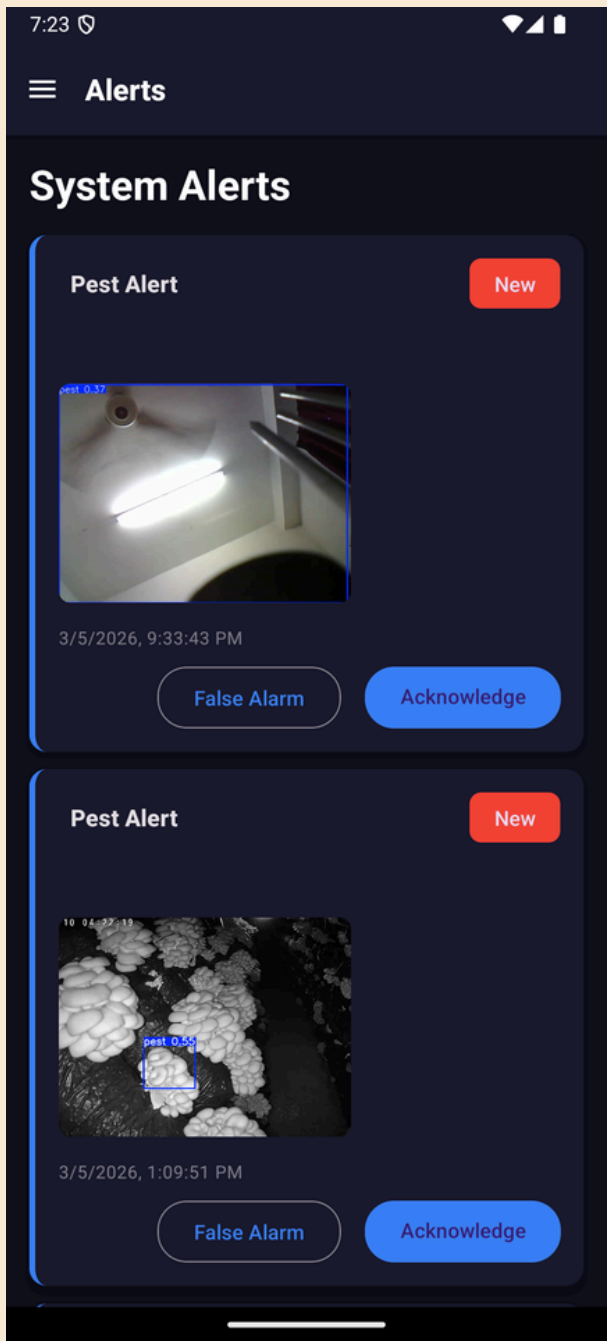
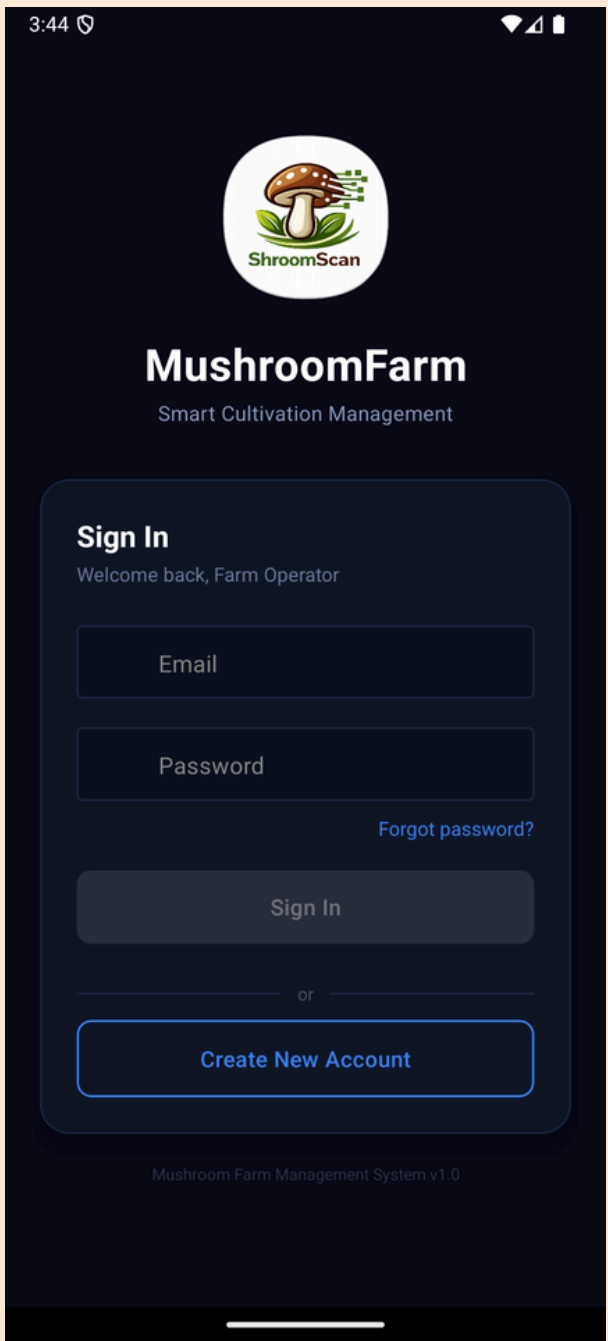
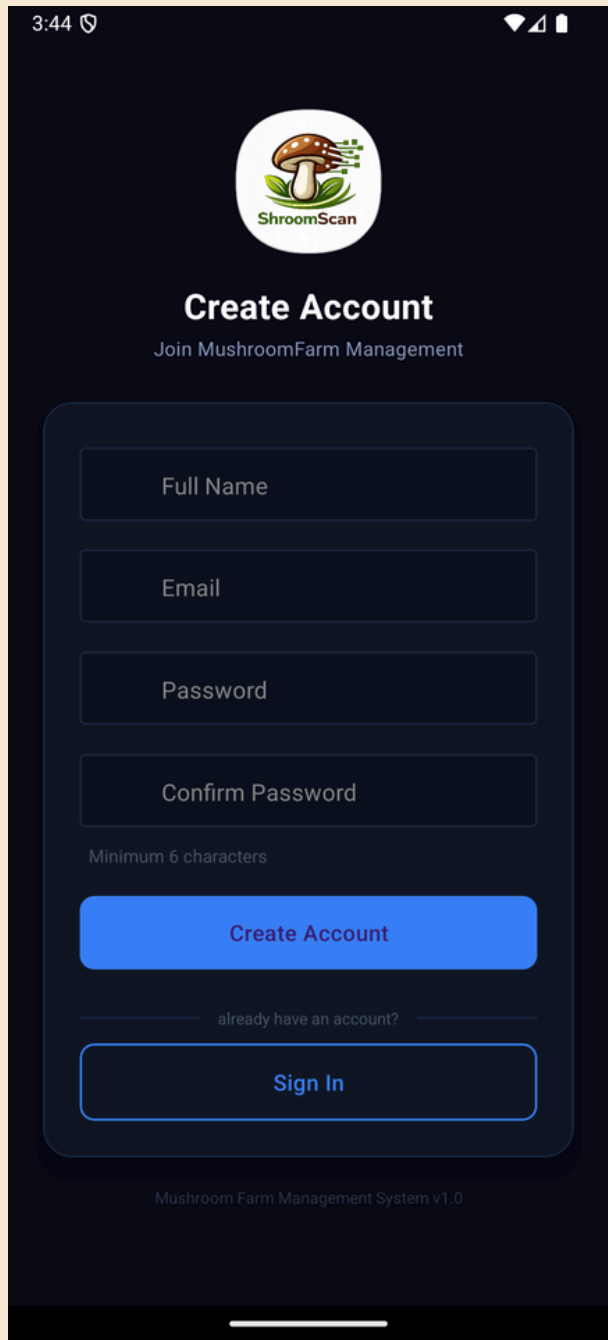
Specialization: Information Technology

# Pest Management Module using Real-Time Detection and Automated Response





# Pest Management Module using Real-Time Detection and Automated Response



# Questions?



# Thank You!